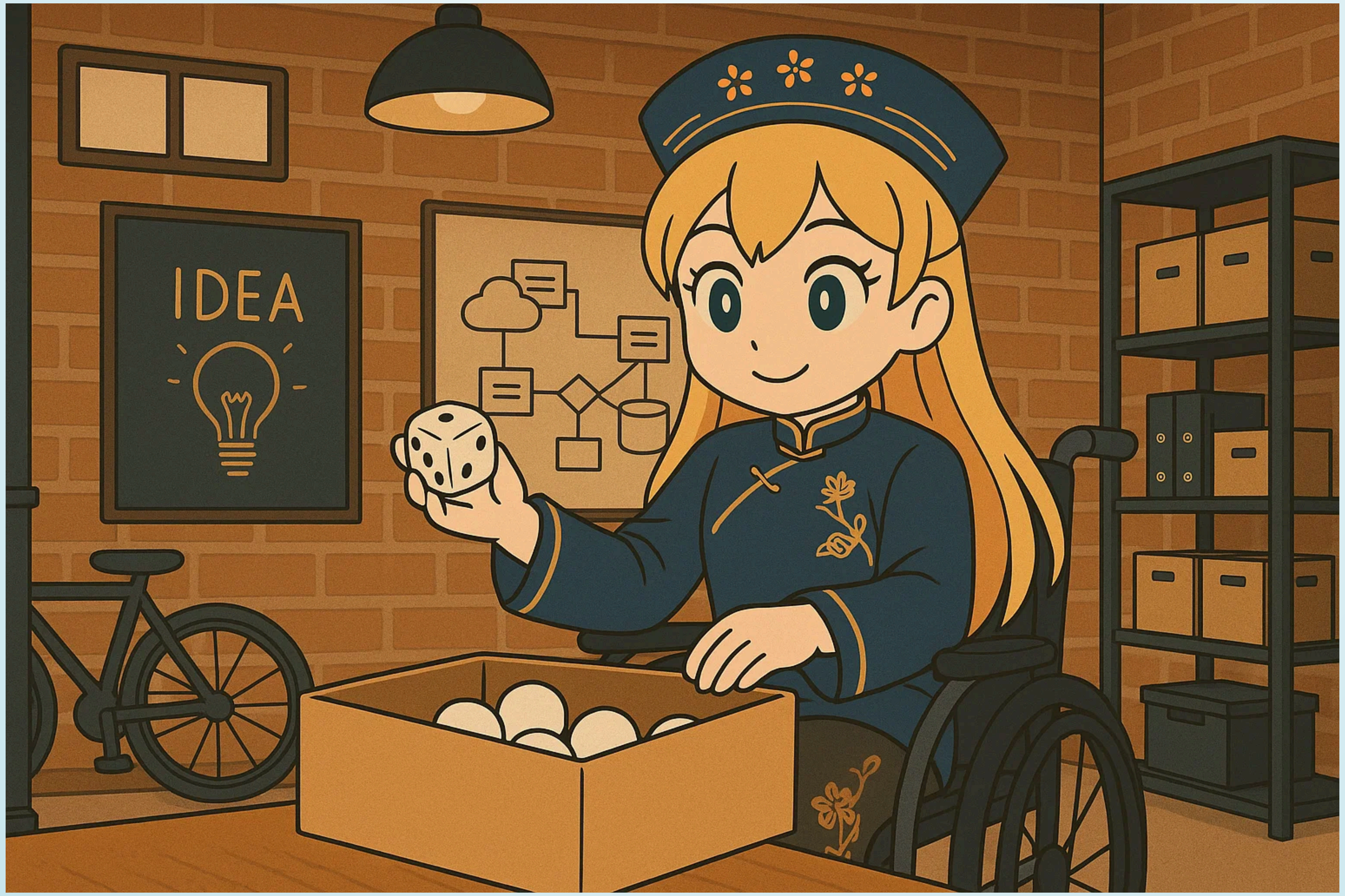
UNDERSTANDING NUCLEAR DECAY AND DIFFUSION PHENOMENA WITH DICE



SELF-INTRODUCTION

- Shark (мег-сск)
  - 🧑‍💻 Freelance Software Engineer
  - 🧑‍🎓 Working student at an online university
- Areas of expertise:
  - 📸 Computer Vision (image recognition/point cloud processing)
  - 🌍 Spatial Information Processing (GIS/remote sensing)
  - ☁️ Cloud Infrastructure Design/IaC (AWS, GCP)
- GitHub

TODAY'S TOPICS

- Consider a simple game where balls are moved based on dice rolls
- Show how this game can reproduce nuclear decay and diffusion phenomena (like the spread of ink stains)!
- **Experience the power of understanding physical phenomena through simple models!**

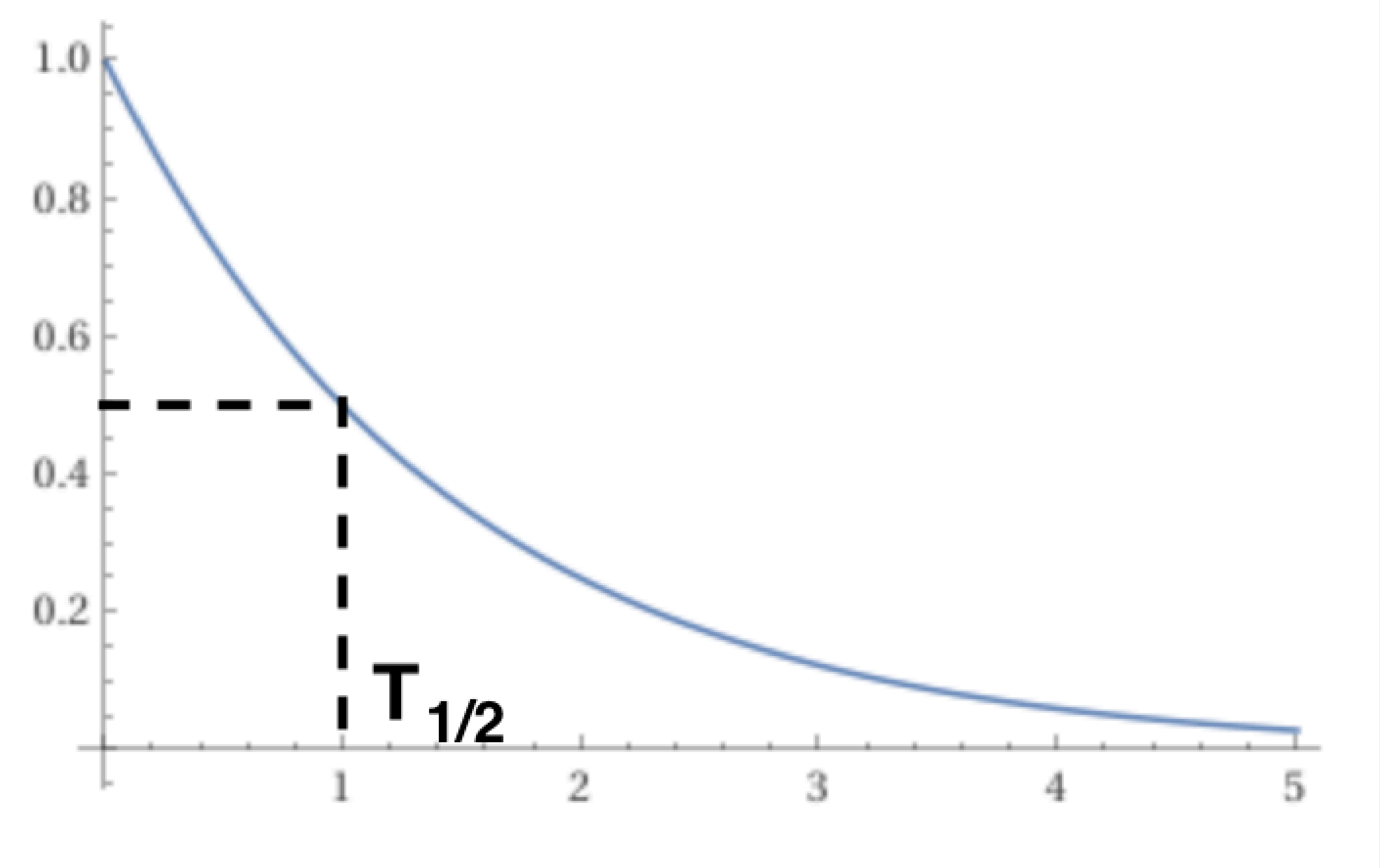
MODELING NUCLEAR DECAY

REVIEW OF NUCLEAR DECAY

- Nuclear decay is a phenomenon where radioactive materials emit radiation
  - Atomic nuclei that emit radiation transform into stable nuclei

RATE OF NUCLEAR DECAY

- The number of radioactive isotopes decreases exponentially: $N(t) = N_0 e^{-\lambda t}$
  - The time it takes for the number of radioactive isotopes to decrease by half is called the half-life $T_{1/2}$



REPRODUCING NUCLEAR DECAY WITH DICE

- A box contains N_0 balls
- Roll a die, and if it shows an even number, remove the ball; if it shows an odd number, leave it in the box
- Perform this operation for all balls

DECREASE IN BALLS AFTER ONE OPERATION

- Probability of a ball being removed: $p = \frac{1}{2}$
- Probability of a ball remaining: $1 - p = \frac{1}{2}$

After one operation, the number of balls will be:

$$N_1 = N_0 - N_0 p = N_0(1 - p)$$

DECREASE IN BALLS AFTER TWO OPERATIONS

- After the first operation, the remaining balls: $N_0(1-p)$
- The same thing happens in the second operation

After two operations, the number of balls will be:

$$N_2 = N_0(1-p)^2$$

REPEATING THE SAME OPERATION

- Since it's the exact same process repeated, after n operations, the number of balls will be:

$$N_n = N_0(1-p)^n$$

GENERALIZING THE DICE GAME

- Let's say we perform this operation λ times with a frequency of Δt
- Define time $t = n\Delta t$
- Set $p = \lambda\Delta t$
  - Our previous example can be considered as the case where $\lambda = 1$ and $\Delta t = 1$

- The number of balls at time t is:

$$N(t) = N_0(1-p)^n = N_0(1-\lambda\Delta t)^n$$

LIMIT OF THE EXPONENT

- Let's recall this formula:

$$\lim_{n\to\infty}\left(1-\frac{x}{n}\right)^n = e^{-x}$$

- Using this formula:

$$N(t) = N_0\left(1-\lambda\Delta t\right)^n = N_0\left(1-\frac{\lambda t}{n}\right)^n$$

REPRODUCING NUCLEAR DECAY

- As $\Delta t \to 0$, taking n so that $t = n\Delta t$ converges:

$$N(t) = N_0 e^{-\lambda t}$$

- This matches the nuclear decay equation!
- We've successfully reproduced nuclear decay with our dice game!

HALF-LIFE

- The half-life $T_{1/2}$ is:

$$N(T_{1/2}) = \frac{N_0}{2} = N_0 e^{-\lambda T_{1/2}}$$

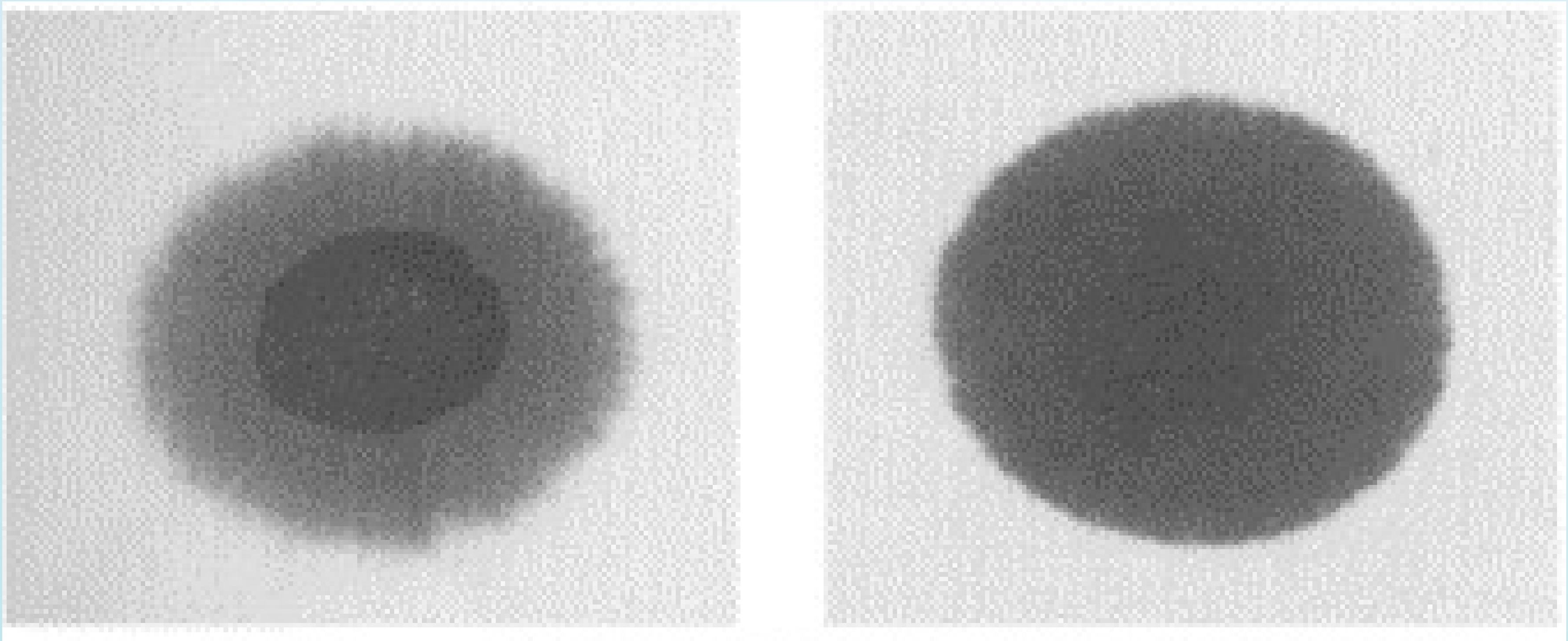
$$T_{1/2} = \frac{\ln 2}{\lambda}$$

- The half-life can be interpreted as "the inverse of the frequency of dice rolls"!

MODELING DIFFUSION PHENOMENA

REVIEW OF DIFFUSION PHENOMENA

- Imagine dropping a drop of ink on a piece of gauze
- Initially, only the area where the ink was dropped is stained
- Over time, the ink diffuses through the gauze



DIFFUSION EQUATION

- Diffusion phenomena are described by the diffusion equation
  - D is the diffusion coefficient
  - ρ is the concentration

$$\frac{\partial \rho(t,x)}{\partial t} = D\frac{\partial^2 \rho(t,x)}{\partial x^2}$$

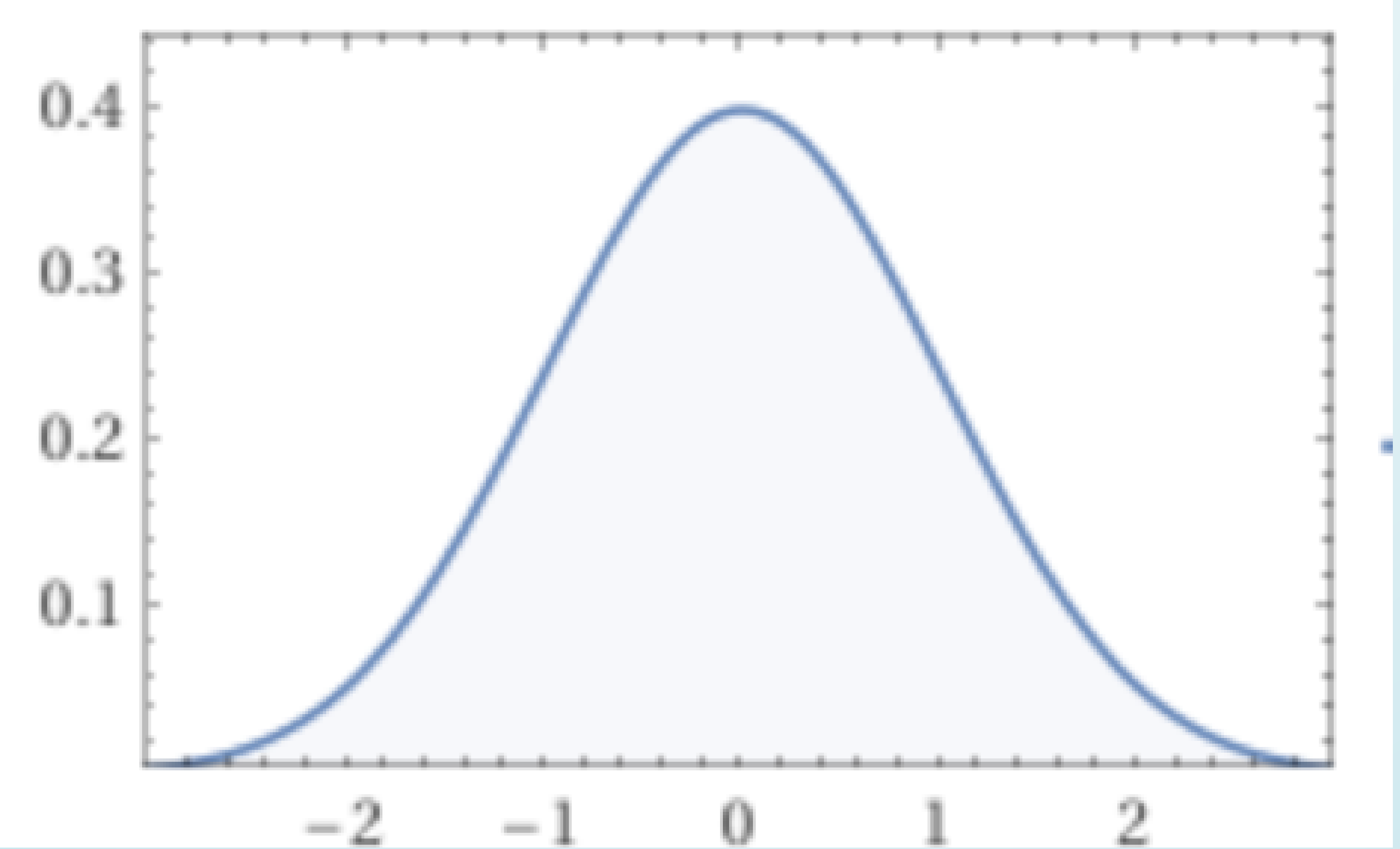
- Solving with the initial condition $\rho(0,x) = \delta(x)$

…

THE SOLUTION OF DIFFUSION

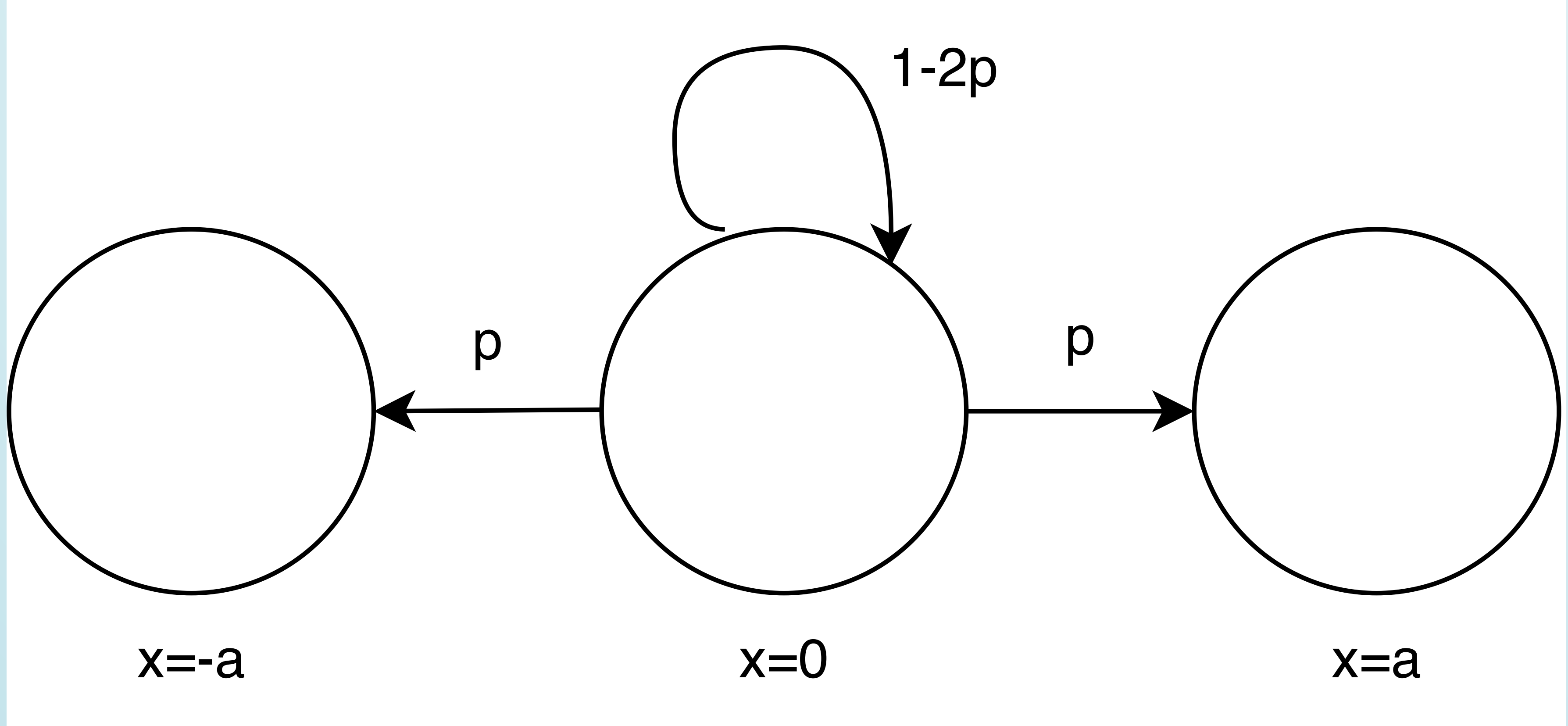
- It is a Gaussian distribution

$$\rho(t,x) = \frac{1}{\sqrt{4\pi D t}} \exp\left(-\frac{x^2}{4Dt}\right)$$



DIFFUSION MODEL WITH DICE

- Move the balls based on the dice roll:
  - If 1 or 2 is rolled, move $+a$ (probability p)
  - If 3 or 4 is rolled, move $-a$ (probability p)
  - If 5 or 6 is rolled, don't move (probability $1 - 2p$)



MOVEMENT OF BALLS AFTER ONE OPERATION

- At $t = 0$, all balls are at the origin $x = 0$
- Perform this operation once for all balls during Δt
- Denote the density of balls as $\rho(t, x)$
- The density after Δt is:

$$\rho(\Delta t, x) = \rho(0, x) - 2p\rho(0, x)$$

$$+p\rho(0, x + a) + p\rho(0, x - a)$$

WHAT HAPPENS WHEN WE REPEAT THE SAME OPERATION?

- Let the density of balls at time t be $\rho(t,x)$
- The density at time $t+\Delta t$ is:

$$\rho(t+\Delta t, x) = \rho(t,x) - 2p\rho(t,x)$$

$$+p\rho(t,x+a) + p\rho(t,x-a)$$

TAYLOR EXPANSION OF BALL DENSITY

- Using Taylor expansion:
  - For the time derivative:

$$\rho(t+\Delta t, x) = \rho(t,x) + \Delta t \frac{\partial \rho(t,x)}{\partial t} + O(\Delta t^2)$$

  - For the spatial derivative:

$$\rho(t, x \pm a) = \rho(t,x) \pm a \frac{\partial \rho(t,x)}{\partial x} + \frac{a^2}{2} \frac{\partial^2 \rho(t,x)}{\partial x^2} + O($$

DERIVING THE DIFFUSION

- Substituting the Taylor expansion into the equation for ball density:
  - Assuming that as $\Delta t \to 0$, the diffusion coefficient converges to $D = \frac{pa^2}{\Delta t}$

$$\rho(t, x) = \frac{1}{\sqrt{4\pi D t}} \exp\left(-\frac{x^2}{4Dt}\right)$$

- This matches the solution to the diffusion equation!
- We've successfully reproduced diffusion

DIFFUSION COEFFICIENT

- The diffusion coefficient D represents the speed of diffusion
- The larger the diffusion coefficient, the faster the diffusion
- The larger the probability p, the larger the diffusion coefficient
- We assumed that as the time interval $\Delta t \to 0$, a behaves such that the diffusion coefficient converges to $D = \lim_{\Delta t \to 0} \frac{pa^2}{\Delta t}$

ADDITIONAL NOTES

- Why do we only consider up to the first-order term for time derivatives?
- And up to the second-order term for spatial derivatives?
- The mathematical rigor is guaranteed by **Itô's formula** (details omitted today)

SUMMARY

- We reproduced physical phenomena with a game where balls are moved based on dice rolls!
  - Nuclear decay
  - Diffusion phenomena
- **We can understand the essence of physics with simple models!**
- There are examples where more complex physical phenomena are understood through simple models
  - I'd like to share some exciting news!

🎉2025 BOLTZMANN MEDAL🎖️

- Professor Yoshiki Kuramoto has been awarded the 2025 Boltzmann Medal!
- The Kuramoto model is applied not only in physics but also in biology, engineering, and social sciences
- The video of metronomes synchronizing is fascinating!

REFERENCES

- A. Murray and I. Hart, The 'radioactive dice' experiment: why is the 'half-life' slightly wrong?, DOI 10.1088/0031-9120/47/2/197
  - A paper explaining how nuclear decay can be reproduced by a dice game with a sufficient number of trials
- Haruaki Tasaki, Brownian Motion and Non-equilibrium Statistical Mechanics
  - Has a very clear explanation of diffusion phenomena through Brownian motion

CALL FOR LT SPEAKERS

- The Physics Meeting is looking for LT speakers!
  - Any genre is welcome!
  - If we don't get any submissions, the organizer will have to do another "LT" (which is actually just a solo recital)...
- If you're interested, please contact us on the Physics Meeting Discord server!

ANNOUNCEMENTS

- The next meeting is scheduled for May 31st
- In addition to LTs, we also welcome suggestions for physics videos on YouTube that you'd like to watch together!